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Swarm intelligence-based green optimization framework for sustainable transportation

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ABSTRACT

Traffic congestion is one of the most critical issues in developing sustainable transportation in smart cities. As the Internet of Things evolves, connected vehicle technology has arisen as an essential research topic in smart, sustainable transportation. This study investigates a decentralized green traffic optimization framework by pushing swarm intelligence into connected vehicles to mitigate traffic congestion. We present a dynamic traffic routing method based on ant species' swarm intelligence for connected vehicles so that they can communicate with each other and their surrounding environment via digital pheromones to perform routing decision-making in a decentralized manner. Traditional pheromones attract other vehicles to move to the optimal path, which will soon be congested if many vehicles travel on that path concurrently. To overcome this limitation, we propose the concept of repelling pheromone, which generates a repulsive force among vehicles so that their travel paths are distributed throughout a road network, resulting in a congestion-free road network. The proposed method is validated in the Simulation of Urban Mobility platform. Simulation findings reveal that the proposed method outperforms baseline methods in mitigating traffic congestion, reducing average fuel consumption and emissions by 13–19% and the average trip duration by 19–28%.

1. Introduction

Due to rapid urbanization, traffic congestion in cities is always a critical issue (Afrin & Yodo, 2020). It leads to higher transportation costs, increasing both fuel consumption and pollutant emissions, causing inestimable impacts on the environment and the health of the people (Tang, McNabola, & Misstear, 2020; Zhang & Batterman, 2013). Effective mitigation of traffic congestion is an essential feature in smart transportation, a vital element of smart, sustainable cities (Bibri, 2018; Silva, Khan, & Han, 2018). Traffic congestion usually occurs when traffic demand exceeds the capacity of the roads. As it is costly to meet traffic demands by solely a higher infrastructure investment, a potential alternative is to improve existing traffic management systems' efficiency. Internet of Things (IoT) interconnects various digital objects through the Internet, creating an intelligent and vast global infrastructure for an information-driven society (Atzori, Iera, & Morabito, 2010). It can be integrated with advanced information and communication technologies such as cloud computing, edge computing, smart grid, and artificial intelligence to realize smart and sustainable cities, which aim to enhance city operations' performance efficiency and the life quality of citizens (Bibri, 2018; Silva et al., 2018). Via IoT, traffic management systems can collect various information on the roads via sensors and give feedback to the vehicles in a centralized manner. However, centralized solutions suffer from two essential issues: scalability, as the centralized server requires high computation and communication when considering large-scale and dynamic traffic networks; and privacy, as the vehicles have to share all information with the centralized server. Therefore, traffic management systems should be operated in a decentralized, self-organized manner.

With recent advances in information and communication technologies such as IoT, edge computing, and vehicular ad-hoc network (VANET), connected vehicles (CV) technology has been emerging to help realize a smart, sustainable transportation system to improve safety, relieve traffic congestion, reduce both fuel consumption and emissions. CV are wireless connectivity-enabled vehicles, which can interact with each other and their surroundings via vehicle-to-everything (V2X) communications, e.g., vehicle-to-vehicle (V2V), vehicle-to-infrastructure (V2I), and vehicle-to-pedestrian (V2P) (Lu, Cheng, Zhang, Shen, & Mark, 2014; Siegel, Erb, & Sarma, 2018). For achieving decentralized and self-organized transportation systems, CV

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technology could take advantage of swarm intelligence, a sub-field of artificial intelligence focused on the collaborative intelligent behavior of natural swarms for problem-solving (Nguyen et al., 2020; Shaikh, El-Abd, Khanafer, & Gao, 2020). For tackling the traffic congestion problem, traffic routing optimization is one of the most promising solutions and has been investigated extensively recently (Del Ser, Osaba, Sanchez-Medina, & Fister, 2019; Jabbarpour, Zarrabi, Khokhar, Shamshirband, & Choo, 2018). Particularly, ant colony optimization (ACO), a well-known probabilistic technique in swarm intelligence, has demonstrated its effectiveness in traffic routing optimization in the CV environment (Bui & Jung, 2019; Cao, Jiang, Zhang, & Guo, 2016; Ho, Lim, Soon, & Chong, 2019; Nguyen & Jung, 2020; Soon, Lim, & Parthiban, 2019; Soon, Lim, Parthiban, & Ho, 2019). ACO is inspired by the social behaviors of real ants that lay down chemical pheromones on their paths as communication media to find out the shortest path from their nest to the food sources (Blum & Dorigo, 2004; Dorigo, Birattari, & Stutzle, 2006). In the ACO-based traffic routing systems, CV are modeled as artificial ants that can exchange traffic information with each other and with road sensors via digital pheromones to find their optimal paths dynamically. However, some of these systems were still deployed in a centralized manner, which cannot handle large-scale and dynamic road networks. Pheromones typically reflect only basic traffic conditions (e.g., traffic density, travel time, travel distance), and a pheromone in an optimal path invites vehicles to select that path for travel. Since many vehicles potentially choose the same path, that path is likely to become congested soon. Also, traffic congestion detection methods have not been included to support the routing decisions, and environmental factors have not been considered. We propose a green traffic optimization framework based on swarm intelligence for sustainable transportation to overcome these shortcomings. The main contributions of this study can be summarized as follows.

- We present a dynamic traffic routing problem considering environmental factors in a CV environment with the objective of trip duration, fuel consumption, and emissions minimization. Unlike traditional traffic assignment problems, this problem considers the dynamicity and uncertainty of the road network and traffic demand.
- We propose a swarm intelligence-based decentralized traffic optimization framework for sustainable transportation in smart cities. We introduce a new type of pheromone in ant species' swarm intelligence, called repelling pheromone, which makes a repulsive force among ants. By pushing this kind of swarm intelligence into CV, the CV can exchange traffic information with each other and with nearby roadside units via digital pheromones and make reroute decisions in a decentralized manner when traffic congestion is detected. With the repelling pheromone model representing occupancy time and fuel consumption rate of CV on the roads, the CV can dynamically select more environmentally friendly travel paths and distribute traffic load throughout a road network, solving the issue of creating different congestion spots during the route guidance, which is a drawback of related works. Thereby, it results in a congestion-free road network, minimizing travel time, fuel consumption, and emissions.
- We performed extensive simulations with a realistic road network on the Simulation of Urban Mobility (SUMO) microscopic simulator. The simulation results demonstrate the efficacy of the proposed method in reducing the average fuel consumption and emissions by 13–19% and the average trip duration by 19–28% in comparison with baseline methods. It is scalable with traffic demand and also works well under different compliance rates and penetration rates.

The remainder of this paper is structured as follows. A literature review is given in Section 2. The repelling pheromone-based traffic optimization mechanism is discussed in Section 3. Simulation settings and outcomes are described in Section 4. Finally, the conclusion is provided in Section 5.

2. Literature review

2.1. Connected vehicles for sustainable transportation

Sustainable transportation solutions are required to meet the increasing demands for mobility in rapid urban developments while mitigating the potentially negative social, economic, and environmental impacts. Traffic congestion, one of the difficult challenges of road transportation in urban areas, often results in undesired traffic delays, increasing energy consumption, producing more air pollutant emissions. Thereby, it causes inestimable impacts on the environment and rising morbidity rates (Tang et al., 2020; Zhang & Batterman, 2013). Traffic congestion can be divided into recurring congestion (caused by predictable events) and non-recurring congestion (caused by unforeseen incidents) (Afrin & Yodo, 2020). The non-recurring congestion is more responsible for traffic delays than the recurring congestion in urban areas due to its unpredictable nature. Effective mitigation of traffic congestion is a crucial challenge in smart transportation and mobility, which is a vital factor in advancing smart, sustainable cities (Bibri, 2018; Silva et al., 2018).

With recent advances in information and communication technologies, CV technology is a rapidly growing paradigm shift toward smart and sustainable transportation (Lu et al., 2014; Siegel et al., 2018). Each CV is envisioned as a smart object with a robust multi-sensor platform, communication capabilities, processing units, and Internet protocol-based networking. CV can reliably exchange the traffic data with the nearby CV through V2V communications and road infrastructure through V2I communications. In particular, once the CV on-board unit assembles traffic data (e.g., location, velocity, source/destination, trajectory, and travel time), it transmits data to roadside units (RSU). Then, RSU further process data to produce additional vehicle kinetics data (e.g., average velocity, acceleration). This data has been explored for congestion detection, incident detection, traffic routing, and energy optimization (Al Mallah, Quintero, & Farooq, 2019; Bui & Jung, 2019; Majumdar, Subhani, Roullier, Anjum, & Zhu, 2020; Nguyen & Jung, 2020).

Smart transportation enabled by CV can substantially enhance safety, mobility, and the environment. Road intersections are particularly vulnerable to massive traffic jams. Data collected from CV and road sensors can be used to control vehicles and intersection signals to maximize vehicle throughput and efficiency (Shaikh et al., 2020; Shi, Li, Han, & Lv, 2020; Xu, Ban, et al., 2018). Xu, Ban, et al. (2018) proposed a cooperative approach of traffic signal control and vehicle speed optimization for CV, which simultaneously optimizes the traffic signal timing and vehicles' speed. Shi et al. (2020) developed a V2X-based multi-intersection coordination approach to maximize the green-wave band of an arterial road in the adjacent intersections. The traffic delay is significantly decreased when signal-free control systems replace traditional traffic lights at intersections under the CV environment. Lee and Park (2012) designed a centralized intersection controller that collects individual CV information and provides the best maneuvers to the CV crossing an intersection to avoid collisions and reduce waiting time, improving air quality and energy savings. Since CV can directly exchange information, several studies proposed decentralized signal-free intersection control systems. CV synchronously negotiate with each other to pass the intersections without a centralized controller or traffic lights (Bui & Jung, 2018; Mirheli, Tajalli, Hajibabai, & Hajbabaie, 2019; Xu, Li, et al., 2018). Besides, traffic congestion can be predicted using historical traffic data from RSU and machine learning to support CV in their routing decisions (Majumdar et al., 2020). The cause of traffic congestion can be elaborated via CV technology in a distributed data mining system (Al Mallah et al., 2019). Lastly, traffic routing optimization, which is also an important task to alleviate traffic congestion, will be strengthened by the CV environment's beneficial information.

2.2. Swarm intelligence

Swarm intelligence is a subset of artificial intelligence and an important class of optimization methods. Swarm intelligence strategies are influenced by collective and self-organizational cooperation between natural agents (e.g., ant colonies, flocking of birds, honey bees, and fish schools), forming decentralized algorithms for problem-solving (Nguyen et al., 2020; Shaikh et al., 2020). Nature inspiration, sociality, and iteration are typical characteristics of the techniques. They are different in the way the agents explore and exploit the search space. Several popular methods include ant colony optimization (ACO), particle swarm optimization, and artificial bee colony. ACO is a typical algorithm in swarm intelligence-based algorithms. It is driven by the foraging behaviors of social ant species, which can discover the shortest way between the nest and food sources by laying down pheromones in their trails (Blum & Dorigo, 2004; Dorigo et al., 2006). Each ant represents a feasible solution to the problem. The solution is probabilistically constructed based on existing pheromone trails and heuristic knowledge available a priori. A shared pheromone table is periodically updated among ants. The better the solution, the higher pheromone concentrations. After a sufficient number of iterations, all ants follow the optimum route with the most pheromone concentrations.

Swarm intelligence is used to solve problems in a variety of fields, such as transportation (Del Ser et al., 2019; Jabbarpour et al., 2018; Shaikh et al., 2020), energy management (Nguyen et al., 2020), recommendation systems (Peška, Tashu, & Horváth, 2019). Several works have recently been carried out using swarm intelligence to develop decentralized control in IoT systems (Zedadra et al., 2018). For example, Nguyen and Jung (2020) proposed an ACO-based traffic routing method, which finds the shortest time travel paths for CV in a decentralized manner. Ma, Hu, Yang, Xu, and Guan (2018) proposed a PSO-driven appliance scheduling algorithm with a day-ahead pricing scheme and solar energy production to minimize energy usage, customer discomfort, and emissions. Bui, Agbehadji, Millham, Camacho, and Jung (2020) presented a dynamic and decentralized ABC-based approach for appliance scheduling to reduce electrical consumption. The appliances are connected and co-operated as a decentralized home energy management system enabled by IoT.

2.3. Traffic routing optimization

A traffic routing problem (also referred to as a traffic assignment problem) is the process of allocating traffic demand to routes on a road network. Wardrop (1952) defined two equilibrium principles for traffic assignment: user equilibrium (users choose routes that result in the lowest individual cost) and system optimum (under social equilibrium, the average travel time is minimized). These concepts can be used to implement routing mechanisms to shift a whole traffic system to the optimum state. In the early stages of solving the traffic routing problems, several studies concentrated on path planning and routing vehicles to their destinations with static conditions, such as route distances and speed limits. For example, the Dijkstra and A* algorithms find the shortest path without considering other road networks' dynamic characters. Google Maps is still dependent on Dijkstra algorithm (Lanning, Harrell, & Wang, 2014). Users can access real-time traffic information, but it could be biased and incorrect because it relies heavily on human feedback. With IoT advancements, traffic control systems can centrally collect various information on the roads and give feedback to the vehicles. Dynamic traffic assignment (DTA) algorithms can approximate user equilibrium with time-varying traffic flows via an iterative simulation procedure (Gawron, 1998). The following are some of the issues that create a substantial difference between theoretical or simulation outcomes and deployable solutions: high computing overhead requirement, tractability for large-scale road networks and traffic volume, real-time guidance capability, effective congestion control, ability to function while certain vehicles are not part of the system or do not obey routing advice. A centralized traffic control system facts two major concerns: scalability and privacy. Since the centralized controller collects the driving information by all vehicles and generates control actions for the entire traffic network, it has high computation and communication requirements, potentially confronting a single point of failure. The routing and rerouting computations should be done at the vehicles in a decentralized, self-organized fashion for solving the issues.

The CV technology can combine with swarm intelligence to realize self-organization, self-adaption, and decentralization in transportation networks. In particular, the ACO, a typical swarm intelligence technique, has been widely applied to solving traffic routing problems (Bui & Jung, 2019; Cao et al., 2016; Ho et al., 2019; Nguyen & Jung, 2020; Soon, Lim, & Parthiban, 2019; Soon, Lim, Parthiban, & Ho, 2019), where each CV is represented as an agent ant interacting with all others via digital pheromones. For example, Bui and Jung (2019) proposed an ACO-based traffic routing mechanism in a CV environment, in which CV deposit their digital pheromone, inciting other CV to move to their paths. CV exchange information and perform routing decision-making in a decentralized manner. Nguyen and Jung (2020) introduced the coloring ants concept to the ACO algorithm to distinguish traffic flows according to different destinations and solve the multi-source multi--destination traffic routing problem. Cao et al. (2016) introduced a pheromone-based traffic management framework, in which traffic pheromones and intention pheromones represent traffic density and future density of the road segments, respectively. Soon, Lim, Parthiban, and Ho (2019) proposed a pheromone-based routing strategy that prioritizes paths with multiple green traffic lights at intersections. Ho et al. (2019) presented a pheromone-based routing scheme in which vehicles can put the pheromones on multi-road segments further away, i.e., segments they intend to travel. A pheromone on an optimal path incites other CV to select that path for travel in existing studies. The systems provide the same path for the many vehicles, potentially creating another traffic congestion, degrading overall traffic efficiency. Unlike previous studies, this study proposes a new type of pheromone, called repelling pheromone, which gives a repulsive force among CV to distribute traffic load throughout a road network. Thereby, the traffic congestion is minimized, promoting sustainable transportation.

3. Green optimization framework for sustainable transportation

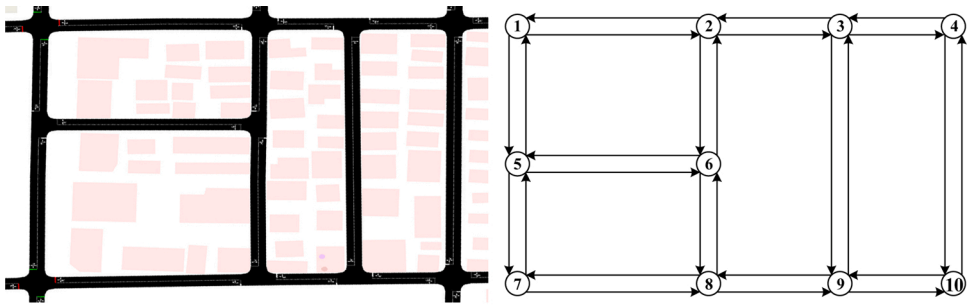
In this section, we present a green traffic optimization problem. Then, a swarm intelligence-based traffic optimization method is proposed. Inspired by ants' swarm intelligence, we model the CV as artificial ants, which interact with each other to mitigate traffic congestion based on a new pheromone model, called repelling pheromone. Finally, a decentralized traffic optimization framework is built to enhance the sustainability of transportation.

3.1. System model and problem definition

A road transport network is defined as a directed graph, whereby intersections are mapped to nodes and roads between them are mapped to edges. The definition of the road transport network is as follows.

Definition 1. (*Road transport network*) The road transport network is defined as a directed graph $\mathcal{G} = (\mathcal{N}, \mathcal{E})$, where $\mathcal{N} = \{i | i = [1, n]\}$ denotes the set of nodes or intersections and $\mathcal{E} = \{(i,j) | i,j \in \mathcal{N}, i \neq j\}$ denotes the set of edges or road segments (two different directions of an actual road are considered as two different road segments). The length of (i,j) is denoted by l_{ij}. $\mathcal{R}$ is the set of roadside units (RSU), where $r_{ij} \in \mathcal{R}$ is the RSU of (i,j).

Fig. 1 illustrates an example of a road transport network and its corresponding directed graph representation. We assume that vehicles entering the road transport network are connected vehicles that can communicate with each other and RSU. The definition of connected



(a) Example road transport network (b) Graphical Representation

Fig. 1. Example of a road transport network and its directed graph representation.

vehicles is given as follows.

Definition 2. (*Connected vehicles*) Letting $\mathcal{K}$ be the set of connected vehicles (CV) that entered the road transport network. Each CV $k \in \mathcal{K}$ has a source node k_s, a destination node k_d, and a start time k_t. The path or route p_k of CV k is defined by a sequence of connected nodes. Given the time-dependent travel costs between nodes, the path with the lowest cost is the shortest path.

There are three types of traffic models: macroscopic, microscopic, and mesoscopic (Hoogendoorn & Bovy, 2001). The microscopic traffic model is selected because it can define the dynamics of individual vehicles in the traffic stream as well as capture interactions among vehicles and between vehicles and road infrastructure. The foundations of microscopic traffic simulation are car-following models, which depict vehicle dynamics when driving in a single lane while maintaining a safe distance from the leading vehicle. We use the Krauss car-following model (Krauß, 1998), which is a time-discrete and space-continuous model for effectively simulating traffic flows. It is a default model in the SUMO simulator (Lopez et al., 2018), which is used to conduct simulations in this study.

In order to measure emissions and fuel consumption of vehicles, Emissions from Traffic (EMIT) model (Cappiello, Chabini, Nam, Lue, & Abou Zeid, 2002) is selected due to its computing efficiency and accuracy. The model first computes the instantaneous tractive power P_{tr} depending on the vehicle velocity k (m/s) and acceleration a (m^2/s) by

$$P_{\text{tr}} = A{\cdot}v + B{\cdot}v^2 + C{\cdot}v^3 + M{\cdot}a{\cdot}v + M{\cdot}g{\cdot}sin\vartheta{\cdot}v \tag{1}$$

where A is the rolling resistance (kW/m/s), B is the speed correction (kW/(m/s)^2), C is the air drag resistance (kW/(m/s)^3), M is the vehicle mass (kg), g is the gravitational constant (9.81 m/s^2), and ϑ is the road grade (degrees). Based on the value of P_{tr}, the fuel consumption rate FR (l/s) is estimated as

$$\text{FR} = \begin{cases} \alpha + \beta v + \gamma v^2 + \delta v^3 + \zeta \text{av}, & \text{if } P_{\text{tr}} > 0 \\ \alpha', & \text{if } P_{\text{tr}} = 0 \end{cases} \tag{2}$$

where α, β, γ, δ, and ζ are constants corresponding to individual vehicles. They were obtained using ordinary least square linear regressions. The EMIT model calculates the engine-out pollutant emissions E (g/s) by

$$E = \lambda + \mu{\cdot}\text{FR} \tag{3}$$

where λ and μ are the greenhouse gas-specific emission index parameters. Among the pollutant emissions products, carbon dioxide (CO_2) is the major production of the complete combustion of fuel and is proportional to the fuel consumption. CO_2 emission values are directly determined from the fuel consumption estimates. Owing to this linear relation, the expressions fuel consumption and CO_2 are interchangeable terms when assessing the environmental effects of vehicles.

Two major issues experienced in the road transport network are traffic signal control and traffic routing. However, traffic signal control is out of the scope of this study. We focus on the dynamic traffic routing problem, particularly on the decentralized green routing method, in which the CV find the least congested, shortest travel paths by themselves and reroute if traffic congestion is detected. In view of sustainable transportation, the objective of the routing method is to find the optimal path with the minimum total fuel consumption, which is the product of the instantaneous fuel consumption rate and trip duration. The objective function of the problem is expressed by

$$\begin{aligned} &\text{minimize} \quad \sum_{i,j,k} x_{\text{ij}}^k{\cdot}\text{FR}_{\text{ij}}^k{\cdot}T_{\text{ij}}^k \\ &\text{subject to} \quad x_{\text{ij}}^k \in \{0,1\}, \quad \forall i,j, i \neq j \in \mathcal{N}, \quad \forall k \in \mathcal{K} \end{aligned} \tag{4}$$

where $x_{\text{ij}}^k \in \{0,1\}$ is a binary variable that is 1 if and only if the CV k passes the road segment (i,j), FR_{ij}^k, and T_{ij}^k are the fuel consumption rate and trip duration of the CV k on road segment (i,j), respectively.

Different from the traditional traffic assignment problems, we consider the dynamicity and uncertainty of real traffic networks and vehicles, i.e., the prior knowledge of the traffic demand or the schedule of the vehicles entering the road network in the future is not given. Therefore, the CV have to adaptively perform routing and rerouting computations to avoid congested road segments without a centralized controller. Fig. 2 shows an example of the dynamic traffic routing process. In the example, the target CV's original intention is to travel to the road segment in front of it. As the RSU on that road detects traffic congestion, the CV can receive a traffic congestion alert message and reroute to an alternative non-congested road segment. As a result, it can avoid the stop-and-go phenomenon on the congested road segment, reducing trip duration, fuel consumption, and emissions. The details of the traffic congestion detection method and decentralized route choice decision-making method based on swarm intelligence are presented in the following sections.

3.2. Swarm intelligence-based green traffic optimization

The green traffic optimization includes a traffic congestion detection method and a green routing method. Once the traffic congestion is detected on the initial route, the green routing method is executed to find an alternative way with more environmentally friendly for CV to travel.

3.2.1. Traffic congestion detection

To detect traffic congestion, we utilize the traffic density of the road segments. The installed infrastructure (e.g., RSU, sensors) serves as a traffic controller for monitoring a particular road segment for detecting traffic congestion in real-time. If a road segment's traffic density is higher than a predefined congestion threshold value, the road segment is congested. The definition of traffic density is given as follows.

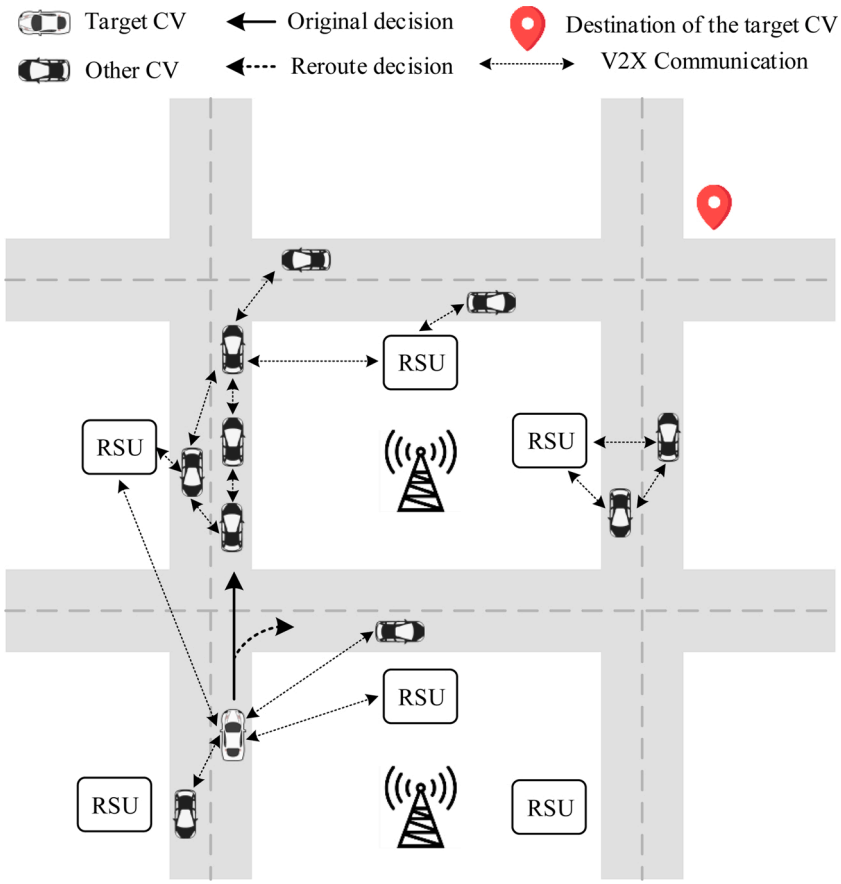


Fig. 2. Overview of the dynamic traffic routing process.

Definition 3. (*Traffic density*) The traffic density of a specific road segment is the rate of the number of vehicles on that road segment vs its capacity. The traffic density D_{ij} of road segment (i,j) at time t is defined by

(5)$D_{ij}(t) = \frac{V_{ij}(t)}{C_{ij}}$where V_{ij} represents the number of vehicles on (i,j), and C_{ij} represents the capacity of (i,j). C_{ij} is estimated as

$$C_{ij} = \frac{l_{ij} \cdot L_{ij}}{\bar{s} + \bar{g}} \tag{6}$$

where L_{ij} is the number of lanes of (i,j), $\bar{s}$ is the average size of the CV, and $\bar{g}$ is the average gap distance between CV.

Based on traffic density, we define an indicator function I_{ij} to determine if road segment (i,j) is congested at time t. The indicator function I_{ij} is formulated as

$$I_{ij}(D_{ij}(t)) = \begin{cases} 1 & \text{if } D_{ij}(t) \geq \phi \\ 0 & \text{otherwise} \end{cases} \tag{7}$$

where $\phi \in [0,1]$ is the predefined congestion threshold value. Determining a suitable value for this threshold is essential for the efficiency of the traffic control system. If it is too low, unnecessary rerouting can occur, leading to a longer trip duration and distance. If it is too high, the rerouting mechanism can be activated untimely, resulting in unavoidable congestion. As suggested by (Pan, Popa, Zeitouni, & Borcea, 2013), the optimal value of ϕ is 0.7 by default.

Thanks to advances in information and communication technologies such as IoT and VANET, the CV can sense their driving environment to provide real-time traffic data using either V2V or V2I communications (Bui & Jung, 2019; Lu et al., 2014; Nguyen & Jung, 2020; Siegel et al., 2018). Each road segment connecting two intersections has a corresponding RSU. When a CV enters a road segment, it broadcasts a message containing basic driving information (e.g., ID, direction, speed, and timestamp). The RSU of the road segment receives the driving data of CV moving on the road segment and then calculates the traffic density. If the RSU detects traffic congestion on the road segment, it will send traffic congestion alert messages to incoming CV, which have not entered the road segment yet, in the adjacent roads. The incoming CV will receive the alert messages and change their initial routes sequentially (first-in-first-out order). When the road segment is no longer under congestion, the traffic congestion alert messages stop being disseminated.

3.2.2. *Repelling pheromone-based routing computation*

In swarm intelligence of ant species, the traditional pheromones on a path of an ant attract other ants to follow and reinforce that path. The same observation is applied to the CV context, where CV send their digital pheromones to the RSU of the path, and the incoming CV will follow the path with the highest pheromone intensity. However, the path will soon be congested if many CV concurrently travel on that path. In contrast, we propose the concept of repelling pheromone as the means for providing traffic information to provide negative feedback for congested road segments. The repelling pheromone value estimated by occupancy time and fuel consumption rate is used to find the alternative paths for CV. The definitions of occupancy time and repelling pheromone are given as follows.

Definition 4. (*Occupancy time*) The occupancy time of a CV is the time required to cover the distance of its size with the current velocity. The occupancy time O_{ij}^{k} of a CV k on road segment (i,j) at time t can be expressed by

(8)$O_{ij}^{k}(t) = \frac{s_k}{v_k(t)}$where s_k is the size of the CV k and v_k is the current

velocity of the CV k.

Definition 5. (*Repelling pheromone*) The repelling pheromone generates a repulsive force among CV, so that they repel each other. The repelling pheromone of a CV k on road segment (i,j) at time t, denoted by τ_{ij}^k, is estimated by the occupancy time and the fuel consumption rate of the CV as

(9)$\tau_{ij}^k(t) = O_{ij}^k(t)\cdot FR_{ij}^k(t)$.The repelling pheromone of the road segment (i,j), denoted by τ_{ij}, is the total repelling pheromone of all CV moving on it. It is updated periodically by

$$\tau_{ij}(t+1) = (1-\rho)\cdot\tau_{ij}(t) + \rho\cdot\sum_{k\in\mathcal{K}}\tau_{ij}^k(t) \tag{10}$$

where $\rho \in (0,1)$ is the rate of pheromone evaporation.

As the occupancy time is affected by the CV's size, it is higher for longer ones, for example, buses and trucks. The initial repelling pheromone value $\tau_{ij}(0)$ on (i,j) is a predefined constant. It is updated periodically according to the update rule of the hyper-cube framework for ACO (Blum & Dorigo, 2004). By definition, the repelling pheromone values are limited to the interval $[0,1]$, and the pheromone evaporation rate controls the strength of the repelling pheromone's influence. We can observe that the more repelling pheromone value on a road segment, the more vehicles traveling on the road segment, and the more vehicles are slowly moving, causing more congestion and fuel consumption.

Based on the traffic congestion indicator function, if congestion is detected on the initial road segment, the CV k obtains alternative K paths from the approaching intersection i by using the K-shortest path algorithm based on known free-flow travel times (Chen, Li, & Lam, 2016; Vanhove & Fack, 2012). The free-flow travel time of a road segment equals the length of the road segment divided by the road segment's speed limit. Then, the CV selects a path based on the repelling pheromone model. The CV should select the path with fewer repelling pheromone, in which the number of traveling CV is smaller as well as the CV are moving faster, resulting in less congestion and fuel consumption. From intersection i, the transition probability $P_{i,p}^k$ for the CV k to select a path $p \in K$ is specified as

$$P_{i,p}^k(t) = \frac{e^{-\tau_{i,p}(t)}}{\sum_{p=1}^{K} e^{-\tau_{i,p}(t)}} \tag{11}$$

where $\tau_{i,p}$ is the total repelling pheromone value of road segments along path p. The CV k can select one of the alternative paths with transition probabilities expressed by the repelling pheromone values. The main idea is to select the path with the highest probability as

$$p = \underset{\forall p}{argmax}\{X\cdot P_{i,p}^k(t)\} \tag{12}$$

where $X \sim \cup([0,1])$ is a uniformly distributed random variable varying from 0 to 1. The random variable provides a better traffic balance and prevents route flapping, which occurs when the congestion moves from one road to another if the same route guidance is followed by a large number of CV.

3.3. Decentralized traffic optimization framework

The traffic control system is designed as a decentralized traffic optimization framework because each component of the system is partly independent and does not have complete global knowledge (Cao et al., 2016; Ho et al., 2019; Soon, Lim, & Parthiban, 2019; Soon, Lim, Parthiban, & Ho, 2019). Accordingly, the system components are defined as agents, including CV agents and infrastructure agents (RSU, traffic light, and traffic camera). All agents exchange information to achieve the objective of alleviating traffic congestion, minimizing trip duration, fuel consumption, and emissions.

The decentralized optimization framework is illustrated in Fig. 3. In particular, real-time data from CV, traffic cameras, and traffic lights are collected by RSU via V2V/V2I communications. An RSU can detect congestion on a specific road segment based on the collected information. If the traffic density is higher than the predefined congestion threshold value, the RSU sends the traffic congestion alert messages to the CV. The CV can then find and move to an alternative non-congested route found by the repelling pheromone model to avoid passing through roads experiencing harsh traffic conditions.

Algorithm 1. Repelling pheromone-based routing algorithm for CV.
Input: $\mathcal{G}$, $\mathcal{K}$, $\mathcal{R}$, ϕ, t.

```
initialization;
while K ≠ ∅ do
    for k ∈ K do
        approaches (i, j) in the current path;
        receives traffic information from r_ij ∈ R;
        detects traffic congestion on (i, j);
        if I_ij(D_ij(t)) = 1 then
            determines alternative K shortest paths;
            P^k_{i,p}(t) = e^{-τ_{i,p}(t)} / Σ_{p=1}^{K} e^{-τ_{i,p}(t)};
            p = arg max_{∀p} {X · P^k_{i,p}(t), X ∈ [0, 1]};
        end
        travels on path p;
        updates its repelling pheromone to new r_ij;
        if k reaches k_dst then
            K = K − k;
        end
    end
end
```

Algorithm 1 presents the repelling pheromone-based routing algorithm for CV. When the CV enters the road network, they find the shortest path for travel with the free-flow travel times as costs. The initial repelling pheromone value of a road segment is also the free-flow travel time value of that road segment. The repelling pheromone information becomes available after some CV already moved on the road network. When the CV reach the end of a road segment, if traffic congestion is detected on their current route plan, the CV will determine alternative paths from the current location using the K-shortest path algorithm with the repelling pheromone values obtained from the neighboring RSU. Then, they select an alternative route using the transition rule. The CV continue to move and update their repelling pheromone values to the RSU on their ways. This cycle is repeated until the CV arrive at their destinations.

Fig. 4 illustrates the workflow from the viewpoint of a CV agent. It is noted that the rerouting procedure is executed when traffic congestion is detected. This differs from the approach of continuously re-evaluating the route of the CV, which wastes computational resources and is what currently happens in existing systems (Bui & Jung, 2019; Cao et al., 2016; Ho et al., 2019; Nguyen & Jung, 2020; Soon, Lim, & Parthiban, 2019; Soon, Lim, Parthiban, & Ho, 2019).

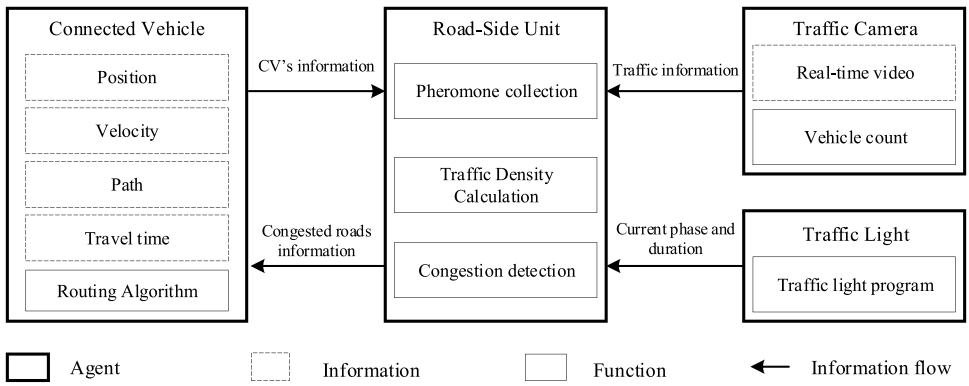


Fig. 3. Decentralized traffic optimization framework.

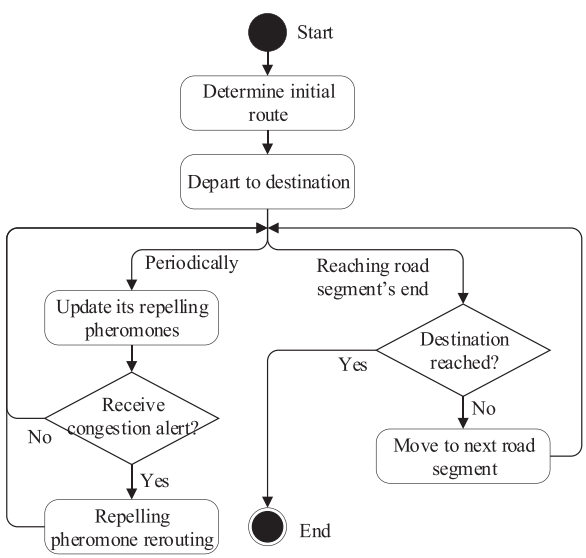


Fig. 4. Workflow from the viewpoint of a CV agent.

4. Performance evaluation

4.1. Simulation settings

To model an urban traffic network and validate the proposed method, we used SUMO,[1] which is a microscopic, time-discrete, and space-continuous traffic simulation (Lopez et al., 2018). It can simulate real-world traffic conditions and decisions of agents. The NETCONVERT and RANDOMTRIPS tools in SUMO are used to generate the road network and vehicle routes.

We downloaded the downtown area road network of Gangnam, Seoul, South Korea from OpenStreetMap[2] for the coordinates of (37.49429, 127.04872) as a case study (Fig. 5). The NETCONVERT tool converts the road network from the OpenStreetMap map to a SUMO map. In particular, Fig. 5a shows the road network of the OpenStreetMap, while Fig. 5b shows the converted road network in SUMO. Besides, Fig. 5c, and Fig. 5d represent road speed limits and the locations of traffic lights in the road network, respectively. The RANDOMTRIPS tool generates vehicles and their source-destination pairs. Python is used to implement the routing methods, monitor performance indicators, perform the routing computation based on real-time traffic data gathered from SUMO via Traffic Control Interface (TraCI, a Python interface to SUMO). We compared the following methods.

- Shortest Time Routing method (STR): Each CV selects the route with the shortest travel time from the source to the destination based on the speed limit. The travel time of a road segment equals the length of the road segment divided by the road segment's speed limit.
- Pheromone-based Routing method (PheR): The fundamental concept of this routing method is to allow CV to select, according to probability, a route marked by higher pheromone concentrations (Bui & Jung, 2019; Cao et al., 2016; Nguyen & Jung, 2020). A pheromone is based on free-flow travel times and traffic density.
- Repelling Pheromone-based Rerouting method (RepelPheRR): This is the proposed routing method based on the repelling pheromone model represented by the occupancy time and fuel consumption rate of CV. It provides the rerouting function during the trip if congestion is detected based on traffic density.
- Dynamic Traffic Assignment method (DTA): The algorithm can approximately achieve user equilibrium via an iterative simulation process (Gawron, 1998). Because of its high computing overhead and need for perfect knowledge of the traffic system, it is not a suitable option for real-time and practical route guidance. However, it can provide the optimal solutions for comparison. The DTA tool implemented in SUMO is used with the number of iterations of 50 as default.

To assess the proposed method's performance, we used four performance metrics implemented in SUMO: average trip duration, average trip distance, average fuel consumption, and average CO_2 emissions, as follows.

- Average trip duration (s): The average travel time taken by all CV from departure to arrival.
- Average trip distance (m): The average distance of the paths traveled by all CV from departure to arrival.
- Average fuel consumption (l): The average fuel expended by all CV between departure and arrival.
- Average CO_2 emissions (g): The average amount of CO_2 emissions emitted by all CV between departure and arrival.

The EMIT model implemented in SUMO is used to estimate the fuel consumption and CO_2 emissions. The CV are randomly generated within a predefined simulation period (3600 s). The main simulation parameters are summarized in Table 1. Two scenarios were conducted to evaluate the performances of the routing methods as follows:

- Scenario with different traffic volumes: To assess the system's scalability, we execute different simulation scenarios with the increasing number of CV from 1000 to 6000.

[1] https://www.eclipse.org/sumo/.
[2] https://www.openstreetmap.org/.

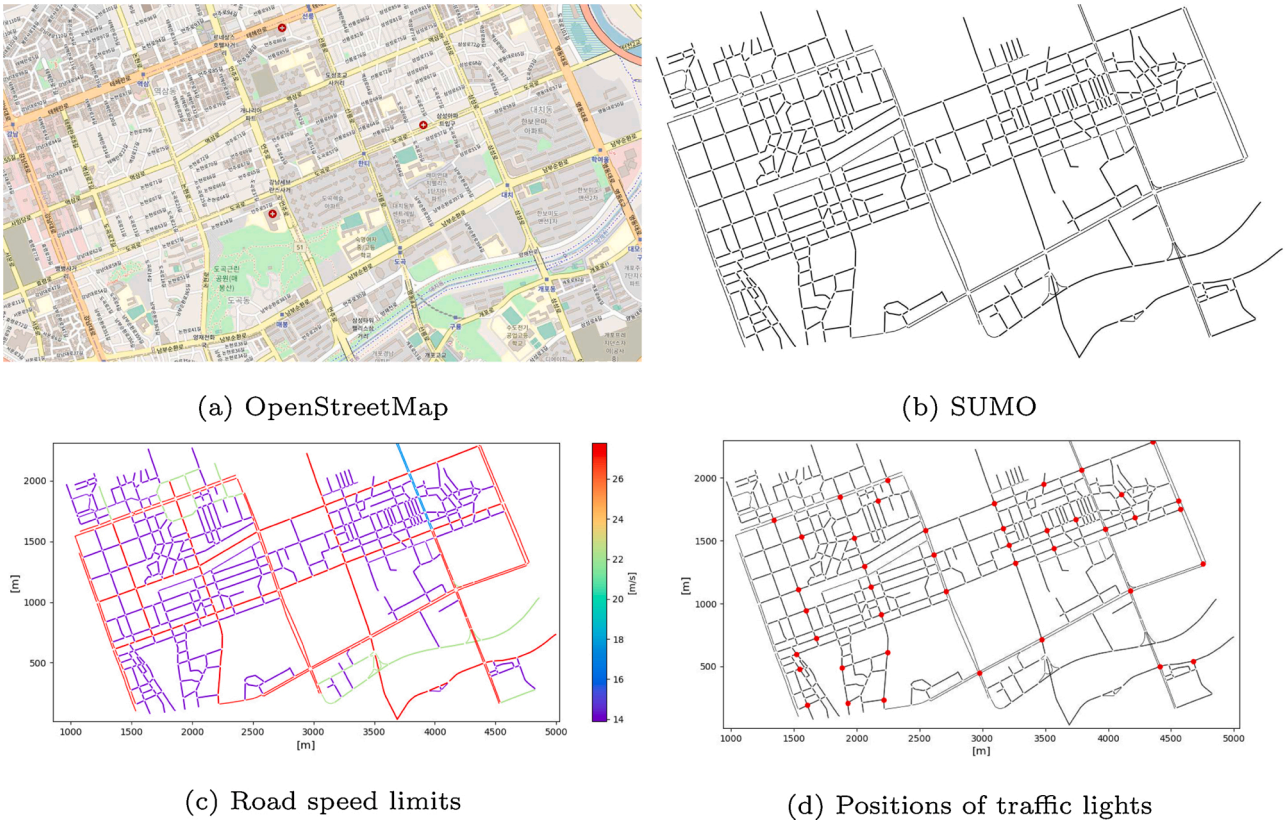


Fig. 5. Road network topology.

Table 1
Road traffic generation parameters.

Parameters	Value
Simulator	SUMO 1.8.0
Road network	Gangnam area, Seoul, Korea
Number of nodes	514
Number of edges	1359
Vehicle generation period	3600 s
Vehicle size	5 m
Vehicle gap	2.5 m
Vehicle emission class	HBEFA3/PC_G_EU4
Traffic light cycle time	90 s
Number of CV	1000–6000
Congestion threshold value ϕ	0.7
Pheromone evaporation rate ρ	0.5

- Scenario with different compliance rates and penetration rates: We assume that all vehicles are CV and fully compliant with the routing method. However, the drivers may not be compliant with the routing method, and the fully CV environment will not be realistic in the near future. The compliance rate is the percentage of vehicles following the route guidance. The penetration rate is the percentage of vehicles sharing their driving information. We test the routing methods in the scenario of 6000 vehicles with different compliance rates and penetration rates, which are 100%, 80%, 60%, 40%, and 20%.

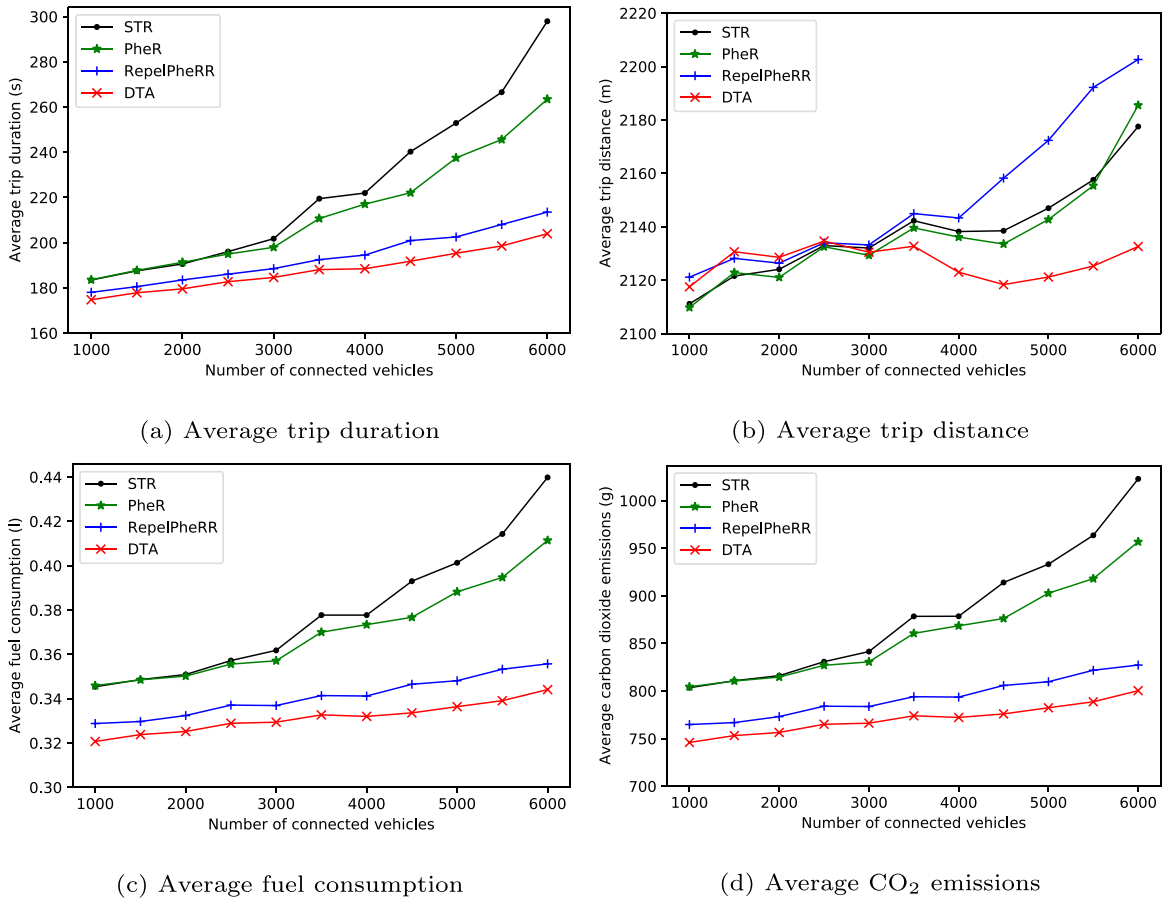
4.2. Results and discussion

4.2.1. Scenario with different traffic volumes

Fig. 6 shows the simulation results of all four routing methods. STR is based on the estimated trip duration for computing the shortest path, while PheR is based on estimated trip duration and traffic density. When the traffic density is low, both methods achieve almost the same results. There is not much difference between the baseline STR and PheR strategies when the number of CV is lower than 3000. As PheR routes CV solely according to pheromone intensity based on traffic density, CV's speed and fuel consumption rate are not well explored. In contrast, RepelPheRR is based on fuel consumption rate and occupancy time, including the size and speed of CV. Therefore, CV using RepelPheRR tend to travel on roads with high speeds of CV and low traffic density to reduce trip duration, fuel consumption, and emissions even these roads may be longer than those used in other methods. As a traffic congestion detection mechanism implemented in RepelPheRR, CV can proactively reroute if traffic congestion is detected in any planned road segments.

Without taking DTA into account, it is observed that RepelPheRR achieves the lowest average trip duration after the simulation is completed, followed by PheR and the baseline STR (Fig. 6a). In other words, the proposed RepelPheRR enables CV to reach their destinations faster than the other two methods. RepelPheRR with the rerouting mechanism causes a slightly higher average trip distance than the other twos (Fig. 6b). However, CV using the RepelPheRR consume less fuel than CV using the other two methods (Fig. 6c). Since CO_2 emissions are directly correlated to fuel consumption, the average CO_2 emissions decrease proportionally with a reduction in the average fuel consumption as expected (Fig. 6d). RepelPheRR not only significantly reduces the average trip duration but also assists CV in traveling on greener paths to promote green transportation. This significant reduction is because CV using RepelPheRR can reroute to non-congested road segments to reduce the frequency of stops in traffic jams. From Fig. 6, with a small trade-off in the average trip distance, RepelPheRR can achieve the results fairly close to the optimal ones of DTA in terms of the average trip duration, average fuel consumption, and average CO_2 emissions.

Table 2 depicts a comparison of the four methods for the scenario of 6000 CV, in which traffic demand is very high, causing heavy traffic congestion. RepelPheRR outperforms PheR in reducing the average trip duration by 18.95%, the average fuel consumption and the average CO_2



(a) Average trip duration (b) Average trip distance

(c) Average fuel consumption (d) Average CO_2 emissions

Fig. 6. Results of the routing methods with increasing traffic volumes.

Table 2
Results for the scenario of 6000 CV.

	STR	PheR	RepelPheRR	DTA
Average trip duration (s)	297.93	263.44	213.53	203.97
Average trip distance (m)	2177.54	2185.50	2202.65	2132.62
Average fuel consumption (l)	0.44	0.41	0.36	0.34
Average CO_2 emissions (g)	1022.84	956.76	827.29	800.42

emissions by 13.5%. It also reported superior results by decreasing the average trip duration by 28.33%, the average fuel consumption and the average CO_2 emissions by 19.1% compared with the baseline STR. Compared with DTA, the proposed routing strategy obtain results near to that of DTA.

Fig. 7 shows the number of moving CV over simulation time for the scenario of 6000 CV. CV are randomly generated throughout the simulation, and they will exit the simulation when they reach their destination. Based on the number of moving CV at a specific time, we can see that the traffic flows using RepelPheRR is smoother than for the other methods except for DTA. The peak number of running CV of RepelPheRR is approximately 400. In contrast, it is around 500 and 600 in the simulations corresponding to PheR and STR, respectively. This is because STR and PheR guide CV to the same optimal paths and do not contain a rerouting mechanism. In contrast, RepelPheRR considers occupancy time and fuel consumption rate of CV for finding optimal routes, and CV also reroute if traffic congestion is detected. It is worth noting that DTA assumes all vehicles in the traffic network prior known to perform routing. In practice, vehicles might emerge at any time, requiring DTA to run its expensive computation repeatedly. Therefore, DTA could be impractical since it cannot find alternate routes quickly enough to alleviate congestion. In contrast, RepelPheRR, which includes traffic congestion detection and proactively decentralized rerouting mechanisms based on the repelling pheromone model, can gradually improve traffic without having prior knowledge of traffic demand.

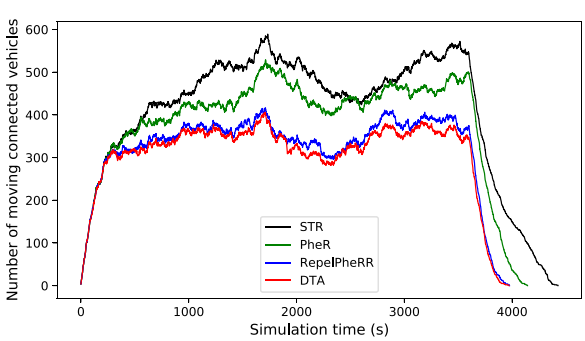
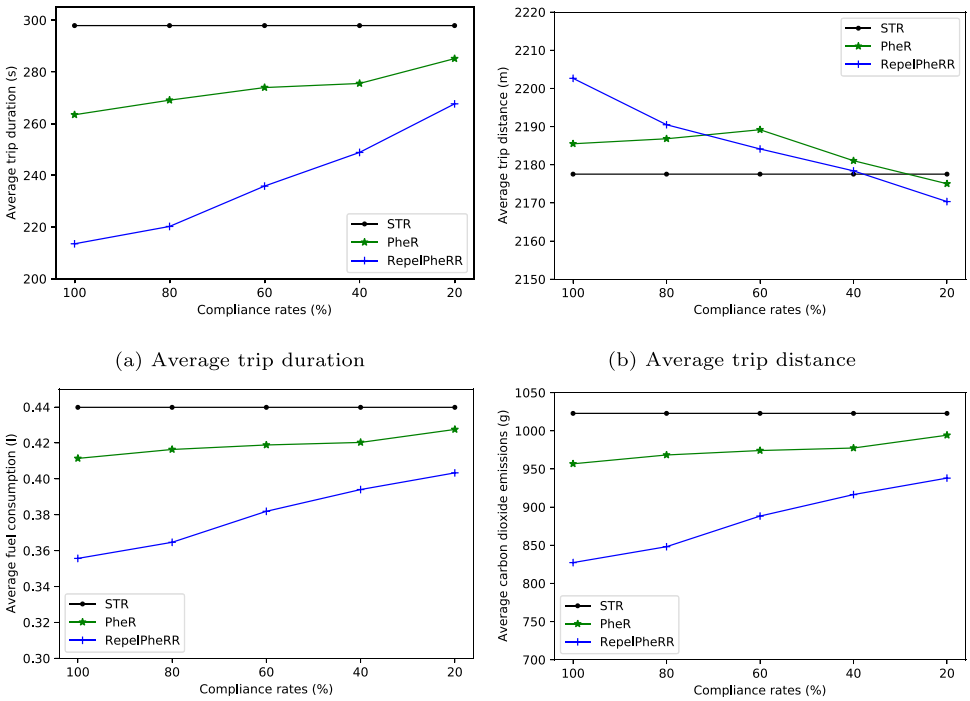


Fig. 7. Number of moving CV throughout the simulation for the scenario of 6000 CV.

4.2.2. *Scenario with different compliance rates and penetration rates*

In this scenario, we did not test DTA since it requires all vehicles to share driving information and follow the route guidance to perform an iterative simulation process to find the user equilibrium.

Fig. 8 shows the simulation results across different compliance rates in the scenario of 6000 CV. Even under low compliance rates, all approaches except STR can still reduce the average trip duration. This is because CV complying with the suggestions can still obtain fairly good routes, which consequently can improve the traffic for the remainder of CV, implicitly decreasing the congestion level on the road network. RepelPheRR outperforms the other approaches tested because it



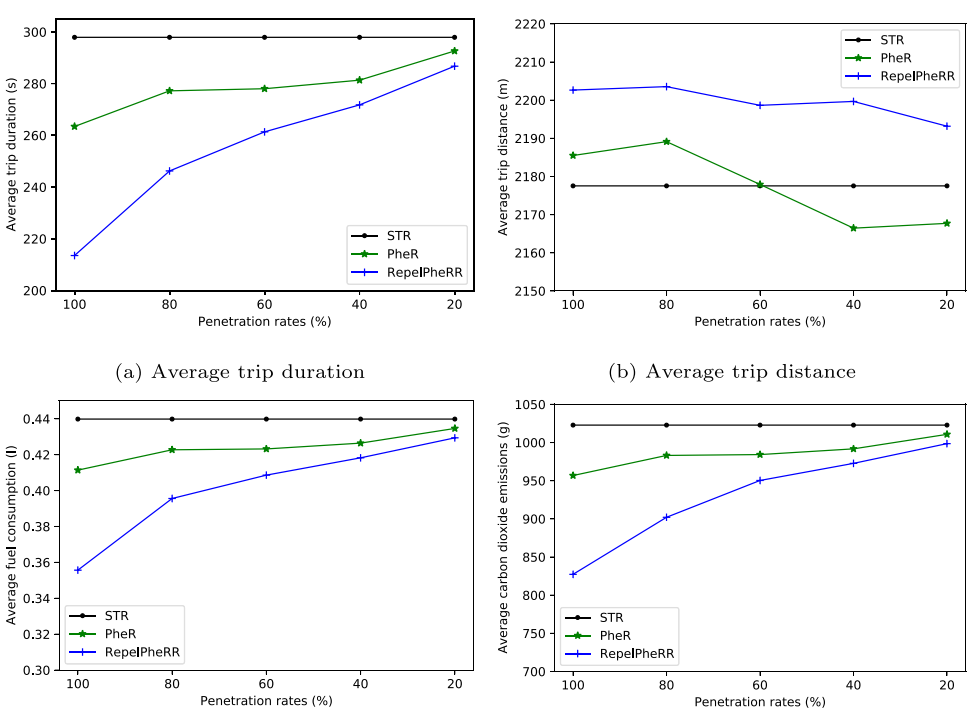
(a) Average trip duration (b) Average trip distance

(c) Average fuel consumption (d) Average CO_2 emissions

Fig. 8. Results for the scenario of 6000 CV under different compliance rates.

integrates traffic congestion monitoring and rerouting mechanisms. The compliant CV using RepelPheRR periodically adjust their routes to avoid congested road segments and avoid competing with the non-compliant CV. As a result, the compliant CV can use better routes, lowering congestion, fuel usage, and air pollutants.

We also performed simulations in a hybrid environment, which is a mix of CV and non-CV. The penetration rate is the proportion of CV that exchange the driving knowledge and follow route guidance. The remaining vehicles are non-CV, which only use STR for routing. Fig. 9 depicts the simulation results with varying penetration rates in the scenario of 6000 vehicles. The effectiveness of the routing approaches except STR increases with the penetration rate. When the penetration



(a) Average trip duration (b) Average trip distance

(c) Average fuel consumption (d) Average CO_2 emissions

Fig. 9. Results for the scenario with 6000 vehicles under different penetration rates.

rate is low (20%), the effectiveness of RepelPheRR is poor and close to that of STR. In this situation, the roadside units' traffic congestion mechanism lacks sufficient data to identify signs of congestion accurately. Once the penetration rate reaches 40%, RepelPheRR can improve the traffic flows and outperforms PheR and STR. For boosting the robustness of the system in the case of low penetration rates, instead of using data collected from CV, traffic congestion can be detected via inductive loop detectors (sensor-based) and cameras (vision-based) (Won, 2020). Thereby, the CV can still receive the traffic congestion alert messages for rerouting to mitigate traffic congestion. The simulation results show that the proposed approach can still improve traffic flows at medium penetration rates.

5. Concluding remarks

In this work, we proposed the swarm intelligence-based green optimization framework for sustainable transportation. Inspired by ants' swarm intelligence, we developed the decentralized routing method based on the repelling pheromone model with the traffic density-based congestion detection mechanism to mitigate traffic congestion in the IoT-based transportation system. The repelling pheromone model depends on the occupancy time and the fuel consumption rate of the CV. The more the pheromone is deposited on a road segment, the worse the traffic flows on that road segment. The CV should avoid congested road segments and reroute to alternative ones found by the repelling pheromone model. We conducted extensive simulations with the realistic road network in the SUMO microscopic simulator. Simulation results showed that the proposed method achieved significant benefits concerning trip duration, fuel consumption, and emissions.

For enhancing the proposed method, several issues need to be considered. First, the studied problem could be extended to consider electric vehicles and investigate the proposed method's impacts on electric vehicles' energy consumption. Second, the proposed method can be integrated with an intelligent intersection control algorithm to reduce traffic congestion further. For example, because the vehicles are assumed to be connected, they can communicate to pass intersections without traffic lights. Another potential future work is to consider advanced traffic prediction techniques for enhancing the effectiveness of the proposed method.

Conflict of interest

The authors report no declarations of interest.

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References

Afrin, T., & Yodo, N. (2020). A survey of road traffic congestion measures towards a sustainable and resilient transportation system. *Sustainability, 12*, 4660. https://doi.org/10.3390/su12114660

Al Mallah, R., Quintero, A., & Farooq, B. (2019). Cooperative evaluation of the cause of urban traffic congestion via connected vehicles. *IEEE Transactions on Intelligent Transportation Systems, 21*, 59–67. https://doi.org/10.1109/tits.2018.2886699

Atzori, L., Iera, A., & Morabito, G. (2010). The internet of things: A survey. *Computer Networks, 54*, 2787–2805. https://doi.org/10.1016/j.comnet.2010.05.010

Bibri, S. E. (2018). The iot for smart sustainable cities of the future: An analytical framework for sensor-based big data applications for environmental sustainability. *Sustainable Cities and Society, 38*, 230–253. https://doi.org/10.1016/j.scs.2017.12.034

Blum, C., & Dorigo, M. (2004). The hyper-cube framework for ant colony optimization. *IEEE Transactions on Systems, Man, and Cybernetics, Part B (Cybernetics), 34*, 1161–1172. https://doi.org/10.1109/TSMCB.2003.821450

Bui, K. H. N., Agbehadji, I. E., Millham, R., Camacho, D., & Jung, J. J. (2020). Distributed artificial bee colony approach for connected appliances in smart home energy management system. *Expert Systems, 37*, e12521. https://doi.org/10.1111/exsy.12521

Bui, K. H. N., & Jung, J. J. (2018). Internet of agents framework for connected vehicles: A case study on distributed traffic control system. *Journal of Parallel and Distributed Computing, 116*, 89–95. https://doi.org/10.1016/j.jpdc.2017.10.019

Bui, K. H. N., & Jung, J. J. (2019). Aco-based dynamic decision making for connected vehicles in iot system. *IEEE Transactions on Industrial Informatics, 15*, 5648–5655. https://doi.org/10.1109/TII.2019.2906886

Cao, Z., Jiang, S., Zhang, J., & Guo, H. (2016). A unified framework for vehicle rerouting and traffic light control to reduce traffic congestion. *IEEE Transactions on Intelligent Transportation Systems, 18*, 1958–1973. https://doi.org/10.1109/TITS.2016.2613997

Cappiello, A., Chabini, I., Nam, E. K., Lue, A., & Abou Zeid, M. (2002). A statistical model of vehicle emissions and fuel consumption. *Proceedings of the 5th IEEE International Conference on Intelligent Transportation Systems (ITSC)*, 801–809. https://doi.org/10.1109/ITSC.2002.1041322

Chen, B. Y., Li, Q., & Lam, W. H. (2016). Finding the k reliable shortest paths under travel time uncertainty. *Transportation Research Part B: Methodological, 94*, 189–203. https://doi.org/10.1016/j.trb.2016.09.013

Del Ser, J., Osaba, E., Sanchez-Medina, J. J., & Fister, I. (2019). Bioinspired computational intelligence and transportation systems: A long road ahead. *IEEE Transactions on Intelligent Transportation Systems, 21*, 466–495. https://doi.org/10.1109/tits.2019.2897377

Dorigo, M., Birattari, M., & Stutzle, T. (2006). Ant colony optimization. *IEEE Computational Intelligence Magazine, 1*, 28–39. https://doi.org/10.1109/MCI.2006.329691

Gawron, C. (1998). An iterative algorithm to determine the dynamic user equilibrium in a traffic simulation model. *International Journal of Modern Physics C, 9*, 393–407. https://doi.org/10.1142/S0129183198000303

Ho, M. C., Lim, J. M. Y., Soon, K. L., & Chong, C. Y. (2019). An improved pheromone-based vehicle rerouting system to reduce traffic congestion. *Applied Soft Computing, 84*, 105702. https://doi.org/10.1016/j.asoc.2019.105702

Hoogendoorn, S. P., & Bovy, P. H. (2001). State-of-the-art of vehicular traffic flow modelling. *Proceedings of the Institution of Mechanical Engineers, Part I: Journal of Systems and Control Engineering, 215*, 283–303. https://doi.org/10.1177/095965180121500402

Jabbarpour, M. R., Zarrabi, H., Khokhar, R. H., Shamshirband, S., & Choo, K. K. R. (2018). Applications of computational intelligence in vehicle traffic congestion problem: A survey. *Soft Computing, 22*, 2299–2320. https://doi.org/10.1007/s00500-017-2492-z

Krauß, S. (1998). *Microscopic modeling of traffic flow: Investigation of collision free vehicle dynamics*.

Lanning, D. R., Harrell, G. K., & Wang, J. (2014). Dijkstra's algorithm and Google maps. *Proceedings of the 2014 ACM southeast regional conference*, 1–3. https://doi.org/10.1145/2638404.2638494

Lee, J., & Park, B. (2012). Development and evaluation of a cooperative vehicle intersection control algorithm under the connected vehicles environment. *IEEE Transactions on Intelligent Transportation Systems, 13*, 81–90. https://doi.org/10.1109/tits.2011.2178836

Lopez, P. A., Behrisch, M., Bieker-Walz, L., Erdmann, J., Flötteröd, Y. P., Hilbrich, R., et al. (2018). Microscopic traffic simulation using sumo. *Proceedings of the 21st IEEE international conference on intelligent transportation systems (ITSC)*, 2575–2582. https://doi.org/10.1109/ITSC.2018.8569938

Lu, N., Cheng, N., Zhang, N., Shen, X., & Mark, J. W. (2014). Connected vehicles: Solutions and challenges. *IEEE Internet of Things Journal, 1*, 289–299. https://doi.org/10.1109/jiot.2014.2327587

Ma, K., Hu, S., Yang, J., Xu, X., & Guan, X. (2018). Appliances scheduling via cooperative multi-swarm pso under day-ahead prices and photovoltaic generation. *Applied Soft Computing, 62*, 504–513. https://doi.org/10.1016/j.asoc.2017.09.021

Majumdar, S., Subhani, M. M., Roullier, B., Anjum, A., & Zhu, R. (2020). Congestion prediction for smart sustainable cities using iot and machine learning approaches. *Sustainable Cities and Society, 64*, 102500. https://doi.org/10.1016/j.scs.2020.102500

Mirheli, A., Tajalli, M., Hajibabai, L., & Hajbabaie, A. (2019). A consensus-based distributed trajectory control in a signal-free intersection. *Transportation Research Part C: Emerging Technologies, 100*, 161–176. https://doi.org/10.1016/j.trc.2019.01.004

Nguyen, T. H., & Jung, J. J. (2020). Multiple aco-based method for solving dynamic msmd traffic routing problem in connected vehicles. *Neural Computing and Applications*. https://doi.org/10.1007/s00521-020-05402-8

Nguyen, T. H., Nguyen, L. V., Jung, J. J., Agbehadji, I. E., Frimpong, S. O., & Millham, R. C. (2020). Bio-inspired approaches for smart energy management: State of the art and challenges. *Sustainability, 12*, 8495. https://doi.org/10.3390/su12208495

Pan, J., Popa, I. S., Zeitouni, K., & Borcea, C. (2013). Proactive vehicular traffic rerouting for lower travel time. *IEEE Transactions on Vehicular Technology, 62*, 3551–3568. https://doi.org/10.1109/TVT.2013.2260422

Peška, L., Tashu, T. M., & Horváth, T. (2019). Swarm intelligence techniques in recommender systems – A review of recent research. *Swarm and Evolutionary Computation, 48*, 201–219. https://doi.org/10.1016/j.swevo.2019.04.003

Shaikh, P. W., El-Abd, M., Khanafer, M., & Gao, K. (2020). A review on swarm intelligence and evolutionary algorithms for solving the traffic signal control problem. *IEEE Transactions on Intelligent Transportation Systems*, 1–16. https://doi.org/10.1109/TITS.2020.3014296

Shi, Y., Li, J., Han, Q., & Lv, L. (2020). A coordination algorithm for signalized multi-intersection to maximize green wave band in v2x network. *IEEE Access, 8*, 213706–213717. https://doi.org/10.1109/access.2020.3039263

Siegel, J. E., Erb, D. C., & Sarma, S. E. (2018). A survey of the connected vehicle landscape-architectures, enabling technologies, applications, and development areas. *IEEE Transactions on Intelligent Transportation Systems, 19*, 2391–2406. https://doi.org/10.1109/TITS.2017.2749459

Silva, B. N., Khan, M., & Han, K. (2018). Towards sustainable smart cities: A review of trends, architectures, components, and open challenges in smart cities. *Sustainable Cities and Society, 38*, 697–713. https://doi.org/10.1016/j.scs.2018.01.053

Soon, K. L., Lim, J. M. Y., & Parthiban, R. (2019). Coordinated traffic light control in cooperative green vehicle routing for pheromone-based multi-agent systems. *Applied Soft Computing, 81*, 105486. https://doi.org/10.1016/j.asoc.2019.105486

Soon, K. L., Lim, J. M. Y., Parthiban, R., & Ho, M. C. (2019). Proactive eco-friendly pheromone-based green vehicle routing for multi-agent systems. *Expert Systems With Applications, 121*, 324–337. https://doi.org/10.1016/j.eswa.2018.12.026

Tang, J., McNabola, A., & Misstear, B. (2020). The potential impacts of different traffic management strategies on air pollution and public health for a more sustainable city: A modelling case study from Dublin, Ireland. *Sustainable Cities and Society, 60*, 102229. https://doi.org/10.1016/j.scs.2020.102229

Vanhove, S., & Fack, V. (2012). An effective heuristic for computing many shortest path alternatives in road networks. *International Journal of Geographical Information Science, 26*, 1031–1050. https://doi.org/10.1080/13658816.2011.620572

Wardrop, J. G. (1952). Road paper. some theoretical aspects of road traffic research. *Proceedings of the Institution of Civil Engineers, 1*, 325–362. https://doi.org/10.1680/ipeds.1952.11259

Won, M. (2020). Intelligent traffic monitoring systems for vehicle classification: A survey. *IEEE Access, 8*, 73340–73358. https://doi.org/10.1109/ACCESS.2020.2987634

Xu, B., Ban, X. J., Bian, Y., Li, W., Wang, J., Li, S. E., et al. (2018). Cooperative method of traffic signal optimization and speed control of connected vehicles at isolated intersections. *IEEE Transactions on Intelligent Transportation Systems, 20*, 1390–1403. https://doi.org/10.1109/tits.2018.2849029

Xu, B., Li, S. E., Bian, Y., Li, S., Ban, X. J., Wang, J., et al. (2018). Distributed conflict-free cooperation for multiple connected vehicles at unsignalized intersections. *Transportation Research Part C: Emerging Technologies, 93*, 322–334. https://doi.org/10.1016/j.trc.2018.06.004

Zedadra, O., Guerrieri, A., Jouandeau, N., Spezzano, G., Seridi, H., & Fortino, G. (2018). Swarm intelligence-based algorithms within iot-based systems: A review. *Journal of Parallel and Distributed Computing, 122*, 173–187. https://doi.org/10.1016/j.jpdc.2018.08.007

Zhang, K., & Batterman, S. (2013). Air pollution and health risks due to vehicle traffic. *Science of the Total ENVIRONMENT, 450*, 307–316. https://doi.org/10.1016/j.scitotenv.2013.01.074